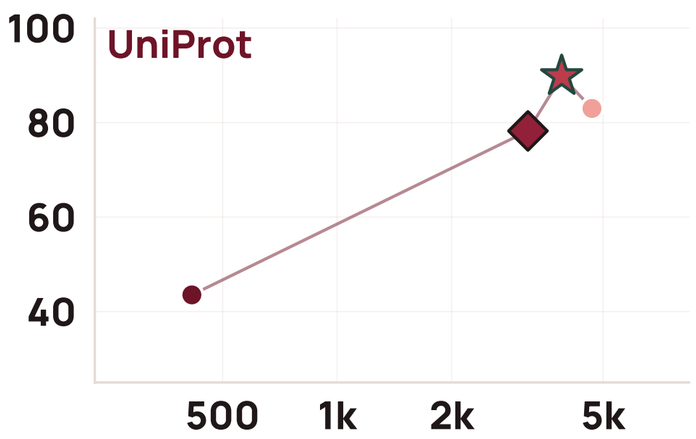
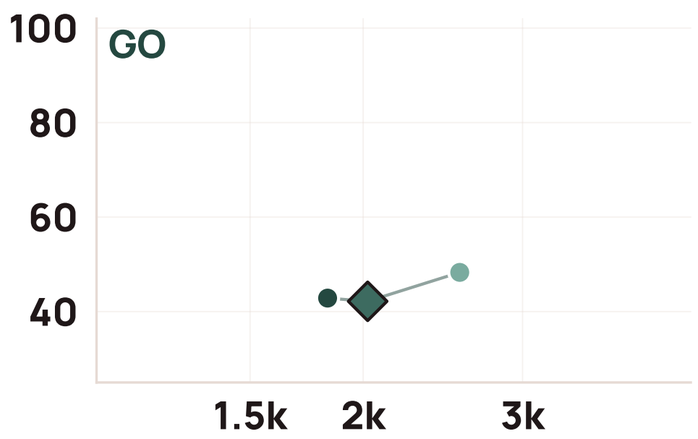
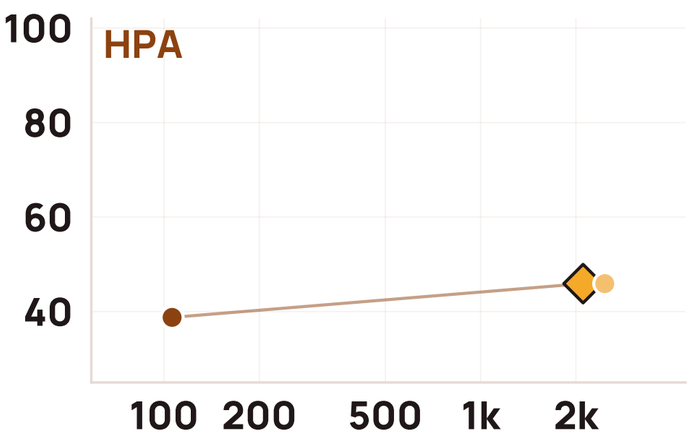
Accuracy on 147-gene benchmark (%)



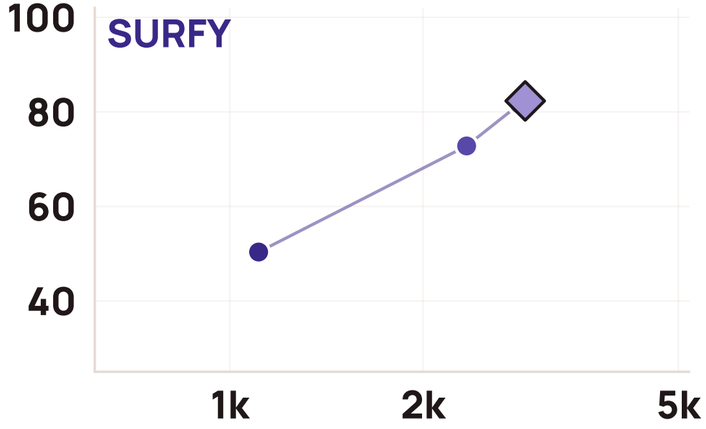
●	strict-4	n= 415	+16/-100
◆	canonical	n=3,175	+70/-96
★	TM+signal	n=3,894	+88/-94
●	permissive	n=4,678	+90/-69



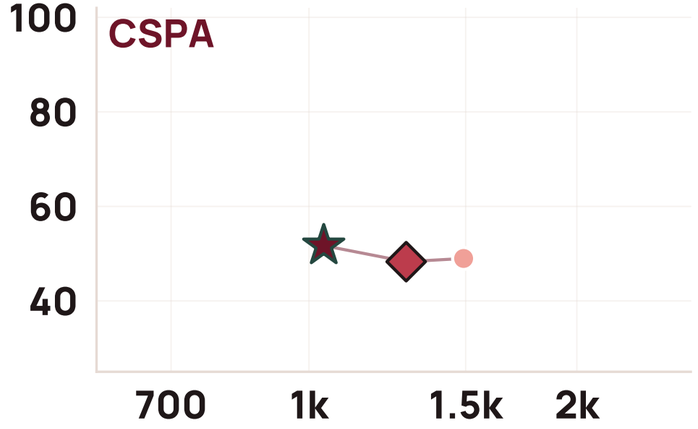
●	exp+curated	n=1,827	+37/-54
◆	canonical	n=2,023	+38/-50
●	+ IEA	n=2,557	+48/-48



●	Enhanced	n= 106	+2/-100
◆	canonical	n=2,107	+36/-62
●	+ Uncertain	n=2,467	+38/-59



●	>0.9	n=1,109	+26/-100
●	>0.7	n=2,339	+62/-96
●	>0.5	n=2,827	+78/-90
◆	canonical	n=2,886	+79/-90



★	HC-only	n=1,039	+38/-79
◆	canonical	n=1,286	+39/-67
●	+ unspecific	n=1,492	+40/-67

Marker shape

- **Alternative cutoff (not used)**
- **Canonical (current merge rule)**
- **Recommended after trade-off audit**

Universe size — proteins this filter would admit (log scale; lower = stricter)